Chapter – 5 User & Group Management

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Understanding Users and Groups

Linux is a multiuser system that relies on accounts—data structures and procedures used to identify individual users of a computer. Managing these accounts is a basic but important system administration skill. Before delving into the details, you need to understand a few basic concepts about user and group administration, which are covered in the following sections. Linux accounts are like accounts on Windows, Mac OS, and other OSs. Some websites use accounts too. Nonetheless, a few details deserve explanation. These include the various Linux users, the nature of Linux groups, and the way Linux maps the numbers it uses internally to the usernames and group names that people generally use.

Learning about Linux User Accounts

Typical Linux accounts are individual user accounts identified via the account’s username. These accounts are for people who need access to the system. Linux accounts can also be accounts for system services called daemons. There can also be specialty accounts that are created for unique purposes. For example, you may want a user account to receive email but not be able to access the local system. Each account requires a unique username. If you can create accounts on your system with mixed-case usernames, Linux treats usernames in a case-sensitive way. Therefore, a single computer can support both ellen and Ellen as separate users. This practice can lead to a great deal of confusion, so it’s best to avoid creating accounts whose usernames differ only in case.

Linking Users Together via Groups

- Following are the three main user administration files:

/etc/passwd

- Keeps user account and password information. This file holds the majority of information about accounts on the Linux / UNIX system.

/etc/shadow

- Holds the encrypted password of the corresponding account. Not all Linux systems support this file.

/etc/group

- This file contains the group information for each account.

/etc/gshadow

- This file contains secure group account information.

Managing Users and Groups

Adding User account and group

To add a new user account and groups, you can run either of the following commands as root.

useradd [new_account]

groupadd [group_name]

useradd Linux

groupadd UTAS

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd Linux
[sudo] password for ubuntu:
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -1
Linux:x:1001:1001::/home/Linux:/bin/sh
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd  UTAS
ubuntu@ubuntu-VirtualBox:~$ cat /etc/group | tail -1
UTAS:x:1002:
```

```
Linux:x:1001:1001::/home/Linux:/bin/sh
```

username → Login name
password → Usually an x, meaning the real encrypted password is stored in /etc/shadow.
UID → User ID number.
GID → Group ID, linked to /etc/group.
comment → Full name or description
home_directory → User's home folder
shell → Login shell (e.g., /bin/bash)

```
UTAS:x:1002
```

1 2 3

1 means Group Name – The name of the group.
2 means Password Placeholder (x)
3 means Group ID (GID) – The unique numerical ID assigned to the group.

Modify or update user id to existing user.

usermod -u 1111 Linux

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -u 1111 Linux
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -1
Linux:x:1111:1001::/home/Linux:/bin/sh
ubuntu@ubuntu-VirtualBox:~$
```

Change the username
(for example existing user as “ Linux” and change this name as “IT”)
#usermod –l IT Linux

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -l IT Linux
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -1
IT:x:1111:1001::/home/Linux:/bin/sh
```

Delete or remove existing user.
#userdel Linux

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd temp
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -2
IT:x:1111:1001::/home/Linux:/bin/sh
temp:x:1112:1112::/home/temp:/bin/sh
ubuntu@ubuntu-VirtualBox:~$ sudo userdel temp
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -2
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
IT:x:1111:1001::/home/Linux:/bin/sh
ubuntu@ubuntu-VirtualBox:~$
```

To add a comment or description for a user, we use the following command.

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd -c "Section-1-Linux" 26J1123
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -1
26J1123:x:1118:1118:Section-1-Linux:/home/26J1123:/bin/sh
```

Adding groups

Creating a new group for read and write access to files that need to be accessed by several users.

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd IT-Network
ubuntu@ubuntu-VirtualBox:~$ cat /etc/group | tail -1
IT-Network:x:1003:
ubuntu@ubuntu-VirtualBox:~$ sudo groupmod -g 2222 IT-Network
ubuntu@ubuntu-VirtualBox:~$ cat /etc/group | tail -1
IT-Network:x:2222:
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG IT-Network IT
ubuntu@ubuntu-VirtualBox:~$ cat /etc/group | tail -1
IT-Network:x:2222:IT
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -2
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
IT:x:1111:1001::/home/Linux:/bin/sh
ubuntu@ubuntu-VirtualBox:~$
```

#groupadd IT-Network

Add group id to group

#groupmod –g 2222 IT-Network

Add Linux user to IT-Network group

#usermod -aG IT-Network IT

groupmod command in Linux is used to modify or change the existing group on Linux system. It can be handled by superuser or root user.
-g, –gid GID: The group ID of the given GROUP will be changed to GID.

Below command will change the group group_old to group_new using -n option.

#groupmod -n group_new group_old

-n, –new-name NEW_GROUP:

The name of group will change into newname.

Delete or remove existing group

groupdel group_name

groups used to display group name for particular

groups user_name

Create two groups named test and temp.

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd test
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd temp
```

Rename the group test to student.

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupmod -n student test
```

Delete the group temp.

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupdel temp
```

Create user named Ahmed.

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd Ahmed
```

Add the user Ahmed to the student group.

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG student Ahmed
```

User Display the groups associated with a specific Ahmed.

```
ubuntu@ubuntu-VirtualBox:~$ groups Ahmed
```

Verify temp group

```
ubuntu@ubuntu-VirtualBox:~$ sudo getent group test
test:x:2224:
ubuntu@ubuntu-VirtualBox:~$
```

To set an expiry date for a user account, we use the following command.

sudo useradd -e date username

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd -e 2025-11-6 Student10
```

Verify the user account expiry date:

chage –l username

```
ubuntu@ubuntu-VirtualBox:~$ sudo chage -l Student10
Last password change                                    : Oct 06, 2025
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : Nov 05, 2025
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
```

The id command in Linux is used to display the user identity and group information for the current or a specified user.

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd 26J2189
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd mid
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG mid 26J2189
ubuntu@ubuntu-VirtualBox:~$ id 26J2189
uid=1128(26J2189) gid=1128(26J2189) groups=1128(26J2189),2228(mid)
```

Viewing Individual Account Records

Display current created users.

```
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd | tail -3
IT:x:1111:1001::/home/Linux:/bin/sh
Student:x:1112:1112::/home/Student:/bin/sh
exam:x:1113:1113::/home/exam:/bin/sh
ubuntu@ubuntu-VirtualBox:~$
```

Display current created user home directory only.

```
ubuntu@ubuntu-VirtualBox:~$ cut -d":" -f6 /etc/passwd | tail -3
/home/Linux
/home/Student
/home/exam
ubuntu@ubuntu-VirtualBox:~$
```

Display current created users name only.

```
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG student Ahmed
ubuntu@ubuntu-VirtualBox:~$ groups Ahmed
Ahmed : Ahmed student
ubuntu@ubuntu-VirtualBox:~$
```

Display current created group name only.

```
ubuntu@ubuntu-VirtualBox:~$ cut -d":" -f1 /etc/group | tail -3
test
exam
Ahmed
ubuntu@ubuntu-VirtualBox:~$
```